

Yash Raj Singh

📍 Kolkata, India ✉ workmail.yrs@gmail.com 📞 +91 91552 77350 in yashhhhh 🌐 rajyashhh 🌐 yashhh.tech

Projects

AI-Powered Natural Language Smart Search for Walmart

[video](#) [github](#)

- Built FastAPI-based AI microservice for multimodal semantic search using Cohere embed-v4.0, indexed in Pinecone.
- Engineered data pipeline processing 1,000 text and 6000+ image vector embeddings with metadata into Pinecone.
- Tools: FastAPI, Pandas, Pinecone, Cohere, React, Tailwind CSS, Express, MongoDB

Y : Full-Fledged Social Media App for College Students

[live](#) [github](#)

- Developed full stack social media platform with secure login, posts & live feeds for college students.
- Launched within university, acquired 150+ users & implemented feedback within 24 hours.
- Tools: React, Tailwind CSS, DaisyUI, Node.js, Express.js, MongoDB, JWT, Vercel

Lung Cancer Model - Under Prof. P.K. Singh

[github](#)

- Achieved **99.54%** accuracy for lung cancer detection using attention-based deep learning model.
- Used ViT-base-patch-16-224 vision-transformer with data augmentation and class weights.
- Tools: TensorFlow, Keras, PyTorch, Scikit-learn, Torch

Work Experience

Founding Engineer – Mandii Plus Pvt. Ltd.

Remote — Aug '25 – Present

- Architected and developed end-to-end truck tracking platform serving 500+ mandi users with real-time GPS monitoring, route optimization, and automated trip management across mobile and web applications.
- Built comprehensive admin dashboard with invoice management system, enabling CRUD operations, PDF generation, media uploads, and analytics integration, reducing administrative overhead by 70%.
- Engineered real-time location tracking APIs using Node.js and Firebase, achieving sub-second data synchronization with 99.9% uptime and supporting concurrent user sessions.

Software Development Intern – House of Developments

Remote — May – Jul '25

- Developed core features for bike rental platform used by thousands of users.
- Managed AWS-based CI/CD pipelines, reducing release time by 40%.
- Enhanced SEO performance through technical optimizations and improved organic traffic.

Education

Jadavpur University

Bachelor of Engineering

Nov '23 - May '27 — CGPA:

7.57/10

British English School

Senior Secondary Education

Apr '23 — 83%

Creane Memorial High School

Secondary Education

Apr '21 — 96%

Position of Responsibility

Entrepreneurship Club Core Member

Kolkata — Apr '24

- Developed event website attracting 100,000+ users and processing 10,000+ registrations.
- Organized multiple entrepreneurship events with 2,000+ participants focusing on innovation.

Sponsorship Team – Srijan

Kolkata — Apr '25

- Secured sponsorship from Canara Bank for annual techno-management fest.

Volunteer – Navprayas NGO

Bihar — May '24

- Engaged with **10,000+** underprivileged students organizing educational events.

Achievements

- Top **10,000** rank out of **260,000** in DSA round of **Adobe's Hackathon**.
- Top **0.8 percentile** out of **180,000** candidates in WBJEE 2023.
- **1st place** "Corporate Devs" & **2nd place** "Data Binge" at E-Weekend.
- **2nd place** ML model training competition with 89% accuracy (IEEE JUSB).

Technologies

Languages: HTML, CSS, C++, JavaScript, TypeScript, Java, Python, C

Frameworks: React, Next.js, Node.js, Tailwind CSS, GSAP, Express.js, FastAPI, Cohere, Pinecone

Developer Tools: Git, GitHub, VS Code, Vercel, Netlify, IntelliJ, Docker, Firebase, Postman